

SEDAR 46 U.S. Caribbean Data Limited Species

Data Workshop

Species Selection:

- Investigate additional data sets and re-evaluate species selection criteria for future stock evaluations.
- For example, consider the information available for queen conch (*Strombus gigas*) in the National Ocean Service's Biogeography visual surveys (Menza et al. 2006) and in data collected by universities in the region.
- Mesophotic reef surveys in western Puerto Rico (García-Sais et al. 2012), visual surveys and passive acoustic monitoring in western Puerto Rico and Mona Island (Scharer-Umpierre et al., 2014), and
- SEAMAP-C (Pagan 2002, Ingram 2014).

To the extent possible, these (and any other datasets) should be integrated and comprehensively summarized to facilitate comparisons and explorations in future analyses.

Life History:

- Representative sampling across size/age spectra for under-sampled US Caribbean stocks.
- Updated studies of life history and demographic characteristics are needed that focus on sampling under-represented size classes, particularly large (old) fishes to provide more accurate estimates of asymptotic length, and small (young) fishes to more accurately estimate the rate at which fishes approach asymptotic length. This recommendation stems from a concern that maximum lengths were too often considerably longer than L_{∞} estimates. This observation could stem from inadequate sampling of the largest length classes, region-specific differences in asymptotic growth (where parameters were borrowed from other regions), or where exploitation has dramatically modified stock structure.
- Additional sampling is also necessary for improving stock-specific maturity schedules, and these data should be fit via modern logistic regressions methods to obtain the most robust estimates of length at maturity.
- Research efforts be put into compilation of various datasets of life history demographic parameters for all exploited species in the tropical western Atlantic, through a Regional Expert Demographic Workshop.

Fishery Statistics:

Commercial research recommendations

- Evaluate the efficacy of existing commercial landings expansion factors used in Puerto Rico; provide recommendations for improved methods to calculate expansion factors;

examine the impact on landings estimates due to methodological changes implemented in 2003 for calculating expansion factors

- Verify, using port samplers or other appropriate methods, self-reported landings in the US Virgin Islands and Puerto Rico
- Obtain species-specific estimates of discards from the commercial sector in Puerto Rico and in the US Virgin Islands
- Quantify the sizes and discard conditions of fish discarded by commercial fisheries in Puerto Rico and in the US Virgin Islands

Recreational research recommendations

- Increase representative sampling of the recreational sector in Puerto Rico and expand to collect recreational data in the US Virgin Islands
- Include spiny lobster and conch in the MRIP in order to estimate recreational catch for these important Caribbean species
- Explore changes in the Puerto Rico recreational catch estimates as a result of the change in intercept protocols and estimation methodologies from MRFSS to MRIP in 2014

Measures of Fishery Abundance:

- Conduct additional examinations to identify auxiliary variables that could be informative in standardization
- Begin the spiny lobster nominal and standardized index further back in time
- Invest in regional scale fisheries-independent surveys to estimate relative (or absolute) abundance
- Investigate methods for subsetting to trips targeting the target species
- Account for change in regulations that may affect CPUE
- Obtain supplementary information and evaluate the use of aggregation of data over gears. The recommendation for SEDAR 46 was to group gear types that were assumed to have similar selectivity's. Additional efforts could help determine when it is or is not appropriate to use gear groups.

Fishing Effort:

- Investigate issue associated with fishers not reporting effort information in St. Croix
- Review any caveats/concerns such as species having more than one dominant fishery or noted changes in fishing behavior
- Extend the data-limited approaches to allow two fisheries, or a single fishery with two distinct types of selectivity/catchability

Length Frequency Distributions:

- The TIP sampling operational framework in Puerto Rico and in the USVI should be reviewed to ensure sampling is representative of the primary fisheries.

- Conduct review of supplemental information on size from data series not readily available for these evaluations.
- Evaluate the use of aggregation of length samples over gears. The recommendation by the SEDAR 46 DW Panel was to group gear types that were assumed to have similar selectivities.
- Address difficulty in assigning the fishing areas to develop a continuous series for the USVI. Develop a consistent time series of area assignments for St. Thomas and St. John. Consider if alternative approaches to aggregating the fishing area information in the TIP data may be feasible.

Assessment Workshop

A number of research recommendations are identified throughout the SEDAR 46 stock evaluation. These arise from the perspective of information content (i.e., data availability, quantity, and quality and information content) and also the modeling approach. Within this context the following discussion and recommendations are made.

Regarding data availability, continued explorations are warranted on the following topics to address uncertainty within key data inputs for data-limited stock assessment models:

1. A statistical review of existing fishery independent surveys to identify an optimum sampling design for development of fishery independent abundance indices. Fishery independent surveys can contribute critical information regarding trends in stock abundance, which can be applied in relatively simple management procedures.
2. Develop indices of abundance for spiny lobster using all available data since 1970s with focus on a fishery independent survey.
3. Investigate more justifiable estimates of stock depletion (Dep) and depletion over time (Dt), such as through Productivity-Susceptibility Analysis (e.g., Cope et al. 2015) or using methods such as mean length estimators.
4. Investigate more justifiable estimates of current stock abundance.
5. Enhanced catch at length by gear sampling is needed to better inform selectivity at age.
6. Investigate fleet dynamics to more accurately capture fishery characteristics.
7. Identify target catch or index levels which could be used in conjunction with catch and index time series.
8. Identify target length levels which could be used in conjunction with catch and a length frequency series.
9. Develop a weighting scheme for length composition and multiple gear fisheries reflective of the stock.

10. Consider organizing species into species complexes for assessment based on similar life history, market characteristics, and vulnerability. This could help streamline the stock assessment process in a data-limited context.

Within the modeling framework used in SEDAR 46, many limitations are acknowledged within an MSE approach. Pragmatically, results are a product of the specific conditions of the simulation, which are assumed to be as simplistic as possible but contain sufficient complexity to reflect the system in a representative way. Methods tend to perform poorly when fundamental assumptions are invalid or inputs are strongly miss-specified. Detecting model misspecification for data-limited scenarios offers additional challenges including evaluating incongruency between data sources. As well, within the implementation model, assumed management target recommendations (i.e., TACs) were taken as catch with no implementation error simulated. Further, no uncertainty was considered in determining TACs via buffers to account for multiple sources of uncertainty (catch reporting, assessment procedure violations, etc). Thus, additional considerations towards confirmation of the stock and fleet subclass components of the operating models explored in SEDAR 46 are warranted. In particular, assumptions regarding the selectivity pattern of fleets should be further examined.

Recommendations for enhancing the practical use of the DLMtool from the analytical team.

- Revisions of the DLMtool software to enhance the model functionality to allow multiple indices of abundance.
- Revision of the DLMtool software to allow age varying M.
- Allow for implementation error of the harvest control rule (e.g., TAC overages) within the implementation model in the MSE.

Recommendations for enhancing the practical use of the DLMtool from the developer (Carruthers (2015a) that the SEDAR 46 analytical team considers of practical relevance to US Caribbean fisheries application of the toolkit:

- Idealized observation models for catch composition data
“Currently, DLMtool simulates catch-composition data from the true simulated catch composition data via a multinomial distribution and some effective sample size. This observation model may be unrealistically well-behaved and favor those approaches that use these data. Harvest control rules must be integrated into data-limited MPs”.

- Harvest control rules
“In the version of DLMtool applied in SEDAR 46 (version 2.1.2), harvest control rules (e.g., the 40-10 rule) must be written into a data-limited MP. There is currently no ability to do a factorial comparison of say 4 harvest controls rules against 3 MPs (the user must describe all 12 combinations). The reason for this is that it would require further subclasses. For example the 40-10 rule may be appropriate for the output of DBSRA but it would not be appropriate for some of the simple management procedures such as DynF that already incorporate throttling of TAC recommendations according to stock depletion.”

- Implementation error

“In this edition of DLMtool there is no implementation error. The only imperfection between a management recommendation and the simulated TAC comes in the form of the MaxF argument that limits the maximum fishing mortality rate on any given age-class in the operating model. The default is 0.8 which is high for all but the shortest living fish species.”

Review Workshop

This section considers the research recommendations initially provided by the DW and AW that were then considered by the SEDAR 46 Review Panel. The Review Panel generally supported the recommendations from the DW and AW, and those from the assessment team. However, the Review Panel extended these recommendations as outlined below. Recommendations fell into two general categories: (1) data; and, (2) model.

Data

One of the fuzziest aspects of the data-limited process was how exactly data reliability was qualified or quantified. We discovered that fishery data precision (e.g., coefficient of variation, CV) was not able to be determined from the current fishery catch sampling methodologies that are employed in the Caribbean. While this was probably a topic of conversation at the DW, there was insufficient discussion of these critical issues in the SEDAR 46 DW/AW report (AW). There needs to be a solid focus on data design strategies as the data-limited process moves forward in the region to establish ACLs for a range of species presently not under consideration.

Thus, two aspects of model inputs must be addressed: (1) life history demographics; and, (2) fishery-dependent data (size-structured catch and fishing effort). Research into what defines the “best” demographic parameters for DLM model inputs, for example, most accurate and precise growth (length-at-age) curve, maximum age (i.e., natural mortality rate), size at first capture (selectivity ogive), size at first sexual maturity (maturation ogive), etc. There seemed to be insufficient attention to these issues in the workshop, and arbitrary (non-estimated) CVs were applied to data inputs. Perhaps the number one priority is to refine the life history demographic parameters identified by the DW across the region, and to improve accuracy and precision of those basic data. This strategy would likely be facilitated by a workshop of technical experts convened, in the near future, to review and analyse existing life history demographic data for all relevant exploited species in the U.S. Caribbean, Southeast U.S. and Gulf of Mexico. When joint parameter variance-covariance is not available, how will estimates of uncertainty for life history demographic parameters, for example, be provided? This would include quantitative justifications for error variances and CVs.

A focus on design-based strategies for ensuring collection of accurate and precise fisheries-dependent commercial and recreational data should be advanced in the region. This would greatly improve fishery-dependent mean (and variance) estimates of landings, discards and the

effort required to obtain them. The sampling protocols must be optimized to ensure representative sampling across size-age spectra over time and space. If precise estimates were obtained in the most recent years, then a data-limited analysis could identify current exploitation rates and resource sustainability. In addition, it makes sense to conduct a statistical review, analysis and optimal sampling design of complimentary fishery-independent surveys as these could provide extremely important spatially-integral, accurate and precise information on exploitation effects by measuring what is left in the water after fishing has occurred.

More work must be done on evaluation of species selection criteria. The adequacy of the choice of species suitable for these pilot species analyses was generally successful. However, a couple of those species provided little guidance on model performance. These analyses revealed issues in three areas: (1) appropriate models and benchmarks; (2) reliable life history demographic data; and, (3) adequate fishery-dependent data.

Model

A review of appropriate data-limited methods should be conducted as soon as possible, under the auspices of SEDAR, to allow evaluation of which methods should really be used in the DLM process for evaluation. Such a technical review would consider: (1) model theoretical basis and assumptions; (2) data requirements; (3) robustness of model to departures from assumptions and data requirements; and, (4) model responses (i.e., biases) to model uncertainty. This would include a systematic analysis of the sources of variability and how they influence OM dynamics. This was nearly impossible to discern in the way that the materials were presented at SEDAR 46, which was no fault of the analysts.

Some of the model estimates produced during SEDAR 46 were very troubling due to either: (1) application of an inappropriate or an inapplicable model(s) or MP; and/or, (2) very wide ranges of error variances, while unknown, that were applied to the input data. As a result, some MPs produced forecasts of unrealistic catch levels, suggesting that their usefulness is highly dubious. Not surprisingly, when appropriate variances and covariances were applied, the median of the output distribution do not change, but the range of model output metrics were substantially reduced. Nevertheless, that did not lead to any material change in the findings of the assessment with regard to MPs that performed better. The argument that this tested the MPs with greater uncertainty and therefore could still be used as a test of robustness was only partially accepted by the review panel.

While this AW was an examination of the potential efficacy of the approach due to its “newness”, and the fact that it was 3rd party application not fully controlled by the analytical team, we believe that in future workshops the analysts should more clearly specify what is desired as an outcome of model simulations, so that the simulations can be more finely tuned to answer specific questions. Generally, feasibility and limitations of MPs to real world

applications is largely determined by data sufficiency and model adequacy. Additionally, there was no guarantee that the sampling algorithms in the OM reflected reality, and to some extent particular methodologies were difficult to assess given the information available to the Panel. In general, the AW would have run more smoothly if more attention were paid to the accuracy and precision of the basic data, and adherence to the assumptions required by the applicable MPs.

A better description and explanation of what is actually going on in the DLMtool OM at the outset would have been useful and clarifying to the Review Panel. As it was, application of methodologies at times appeared quite *ad hoc*, particularly as related to application of means, variances and coefficients of variations of model parameters. The parameters were treated as independent random variables, when we know they are dependent. But this is in fact the DLMtool default as it tries to cover a very wide range of uncertainties. There were a number of unclear definitions, such as “model stability”, which roughly translated to how many simulation runs were required for an input level of variation where for some unspecified reason, all model parameters seemed to be varying simultaneously. This would suggest that some further attention to model sensitivity is highly warranted. Concepts as straight-forward as the number of required model runs to achieve stationarity were not well substantiated.

The apparent uncertainty in both data and models for U.S. Caribbean species suggests caution when selecting MPs intended to provide management advice. Selection of a particular MP for providing catch allocation strategies for management should consider: (1) MP sensitivity to parameters; (2) satisfying model assumptions; and, (3) information quality.

Recommendations

More precise and clearer descriptions and rationales for model thresholds and benchmarks used in the DLM process are needed. Analyses presented at the AW focused heavily on fishery yields (i.e., catches) which made it difficult to discern the rationale for what constituted a particular preferred choice of the MPs. A broader perspective might be entertained when setting OFLs and other appropriate benchmarks. This would likely include yield risks as they relate, in addition, to benchmarks specific to both economic and ecological risks. Adherence to this philosophy would require that model thresholds are set at more conservative resource use levels than are presently considered, and this in turn would avoid theoretical searches of infeasible or impractical model decision space. It is probably not useful to go too far into the weeds in trying to assess the full complexity of a fishery at first, rather the assessment needs to focus on distinguishing sustainable from non-sustainable rates of exploitation, and then identify the appropriate annual catches required to sustain the resource(s). If multiple MPs or a subset of tools are used, then some consideration must be given to model averaging. It would appear from the AW that many of the proposed estimation methods and MPs are non-starters from the outset. This seems an opportune time to conduct a thorough analysis of DLMtool efficacy. The Panel feels that the approaches presented could have broad potential for use in the Caribbean, but still

require deeper, more thoughtful consideration to determine what avenues of application allow one to achieve the greatest utility of the tool.